Module 11: Genomics and Biotechnology — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Explain how restriction enzymes and ligase create recombinant DNA
2. Describe the three steps of PCR
3. Explain how gel electrophoresis separates DNA by size
4. Describe how DNA fingerprinting works
5. Explain what GMOs and gene therapy are
6. Describe what CRISPR is and why it matters

The Big Picture

> Biotechnology takes what we know about DNA and puts it to work. We can now copy DNA, cut and paste genes, identify criminals, treat genetic diseases, and edit the genome itself.

Key Terms

Cutting and Pasting DNA

- **Restriction Enzyme** — Molecular scissors that cut DNA at a specific sequence
- **Sticky Ends** — Overhanging single-stranded ends that can pair with matching fragments
- **DNA Ligase** — Molecular glue that seals DNA fragments together
- **Recombinant DNA** — DNA made from two different sources spliced together
- **Plasmid** — A small circular DNA in bacteria; used as a delivery vehicle (vector)
- **Transformation** — Putting a plasmid into a bacterial cell

PCR (Copying DNA)

- **PCR** — Makes millions of copies of a DNA segment
- **Step 1: Denature (~95°C)** — Heat separates the two DNA strands
- **Step 2: Anneal (~55°C)** — Short primers stick to the target region
- **Step 3: Extend (~72°C)** — Taq polymerase builds new strands
- **Taq Polymerase** — A heat-proof enzyme from hot spring bacteria
- After each cycle, DNA doubles: 1 → 2 → 4 → 8 → 16... (2^n after n cycles)

Gel Electrophoresis (Sorting DNA)

- DNA is **negative** (phosphate groups) → moves toward the **positive** end
- **Small fragments go farther**, large fragments stay near the top
- **DNA ladder** — Known-size fragments used as a ruler

DNA Fingerprinting

- **STR (Short Tandem Repeat)** — A short DNA sequence repeated different numbers of times in different people
- Everyone (except identical twins) has a unique STR pattern
- Used in: **forensics** (crime scenes) and **paternity tests**
- In a paternity test: the child's bands that don't match mom must come from dad

Genetic Engineering

- **GMO (Transgenic organism)** — Has a gene from another species
- **Example: Insulin** — Human insulin gene put into bacteria → bacteria make insulin
- **Example: Golden Rice** — Beta-carotene gene added to rice → helps prevent vitamin A deficiency

Gene Therapy and CRISPR

- **Gene Therapy** — Treating disease by giving a patient a working copy of a broken gene
- **Ex vivo** — Cells taken out, fixed, put back
- **In vivo** — Treatment delivered directly into the body

- **CRISPR-Cas9** — A gene-editing tool that can cut, delete, or fix specific genes
- **Somatic therapy** — Changes body cells only (not inherited)
- **Germline editing** — Changes egg/sperm/embryo (IS inherited — raises ethical questions)

Why It All Works

The genetic code is **universal** — the same codons mean the same amino acids in bacteria, plants, and humans. That's why a bacterium can read a human gene and make a human protein.

Study Tips

1. **Know the tools** — Restriction enzymes CUT, ligase GLUES, polymerase COPIES, gel SORTS
2. **PCR = photocopier** — Three temperatures, three steps, exponential growth
3. **Small = far in gel** — Small fragments travel farther
4. **STRs for ID** — Like a DNA barcode unique to each person
5. **CRISPR = find and replace** — Like using find-and-replace in a document, but for DNA
6. **Universal code = biotech is possible** — Same language in all life